

PostgreSQL Setup on VPS (Production Ready)

1. Install PostgreSQL

```
sudo apt update
sudo apt install -y postgresql postgresql-contrib
```

2. Secure PostgreSQL

```
sudo -u postgres psql
ALTER USER postgres WITH PASSWORD 'YourStrongPasswordHere';
\q
```

3. Create Database & User

```
sudo -u postgres psql
CREATE USER appuser WITH ENCRYPTED PASSWORD 'app_password';
CREATE DATABASE appdb OWNER appuser;
GRANT ALL PRIVILEGES ON DATABASE appdb TO appuser;
\q
```

4. Allow Firewall Access

```
sudo ufw enable
sudo ufw allow 5432/tcp
sudo ufw allow 'OpenSSH'
```

5. Enable Remote Access

```
sudo nano /etc/postgresql/16/main/postgresql.conf
# set:
listen_addresses = '*'

sudo nano /etc/postgresql/16/main/pg_hba.conf
# add:
host all all 0.0.0.0/0 scram-sha-256

sudo systemctl restart postgresql
```

6. Connection Commands

```
psql -d appdb -U appuser
psql -h your_vps_ip -p 5432 -d appdb -U appuser
```

7. Connection URI

```
postgresql://appuser:app_password@localhost:5432/appdb
```

8. Creating Another Database

```
sudo -u postgres psql
CREATE USER newappuser WITH ENCRYPTED PASSWORD 'new_password';
CREATE DATABASE newappdb OWNER newappuser;
GRANT ALL PRIVILEGES ON DATABASE newappdb TO newappuser;
\q
```